Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 5 – Chemistry of Food

Marking Scheme

Date: Wednesday September 18th, 2019

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	47	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task…

- Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.
- You may keep the data book.

Electronic Devices

Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

***Q1.* (7 marks)**

a) Place a number (1-4) in front of the following types of fatty acids of similar chain length to indicate their expected melting point where 1 is the lowest melting point. (2 marks)

Number 1-4	Type of fatty acid
1	poly-unsaturated
2	cis-mono-unsaturated
3	trans-mono-unsaturated
4	saturated

½ mark each correct

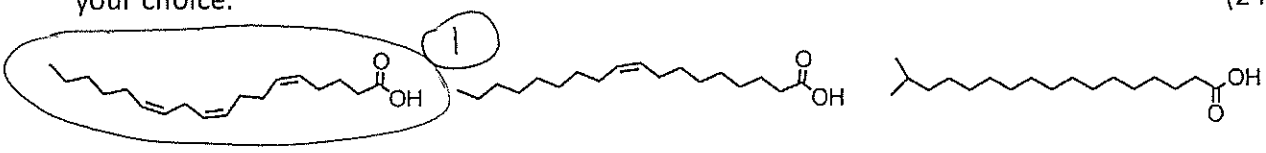
b) Use the data book to provide the name of one example of a mono-unsaturated fatty acid. (1 mark)

palmitoleic acid or oleic acid (1)

c) How many mole of I_2 molecules would you expect to react with one mole of the arachidonic fatty acid? Explain your answer. (2 marks)

Arachidonic fatty acid has 4 C=C. (1)
Each will react with one I_2, so it will take 4 mole of I_2 to saturate it. (1)

d) Of the three fatty acids shown here, which is **most** prone to oxidative rancidity? Circle and explain your choice. (2 marks)



C=C are most likely to be oxidised, fatty acid circled has the most (3) C=C (1)

***Q2.* (5 marks)**

Bioethanol can be made from the cellulose, $H\text{-}(C_6H_{10}O_5)_n\text{-}OH$ in wood pulp.

a) The first step in the production of bioethanol is the hydrolysis of cellulose to produce glucose.

Write a balanced equation for the hydrolysis of cellulose to produce glucose. States are not required. (1 mark)

$H(C_6H_{10}O_5)_n\text{-}OH + (n-1)H_2O \rightarrow nC_6H_{12}O_6$ (1)

b) What is the name of the functional group broken by the hydrolysis reaction? (1 mark)

ether or glycosidic link. (1)

c) The second step of the production of bioethanol is the fermentation of glucose.

Write the balanced equation for the fermentation of glucose. (1 mark)

$C_6H_{12}O_6\ (aq) \rightarrow 2C_2H_5OH\ (aq) + 2CO_2\ (g)$ (1)

states not required.

d) 1.30 g of the glucose produced (M = 180 g mol^{-1}) underwent complete combustion. The energy released was used to heat an unknown mass of water. If the temperature of the water increased by 24.3 °C and it is assumed no heat was lost, calculate the mass of the water heated.

(2 marks)

$q = 16000 \times 1.30$
$= 20,800$ J (1)

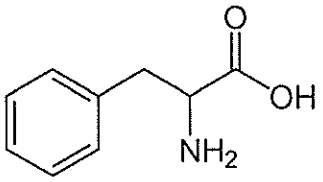
$m = q / c\Delta T$
$= \frac{20800}{4.18 \times 24.3}$
$= 205$ g (1)

−1 if unit not included.

Q3. (6 marks)

Phenylalanine is an amino acid found in a number of foods, including meat, fish, some fruits, cottage cheese, lentils, peanuts and sesame seeds.

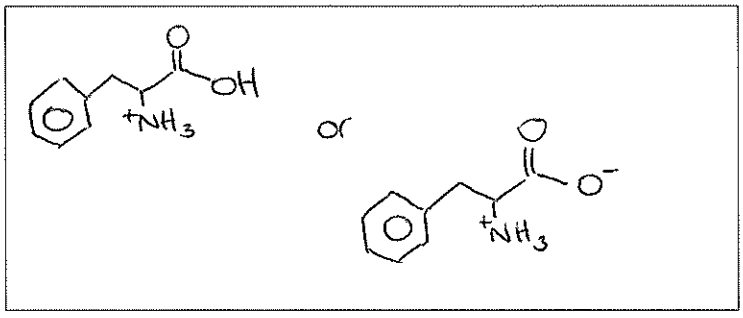
The structure of Phenylalanine is shown here:



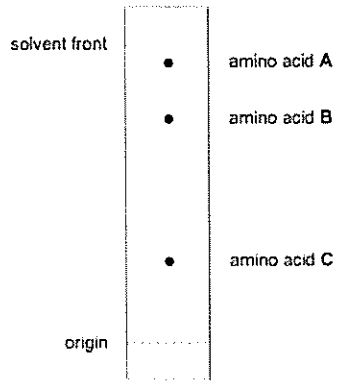
a) Identify two features of the structure of phenylalanine that classify it as an amino acid. (2 marks)

2 of: carboxy and amine functional groups.
organic side chain

b) Phenylalanine is found in tomato juice, which is acidic. Draw one possible structure of phenylalanine in tomato juice. (1 mark)



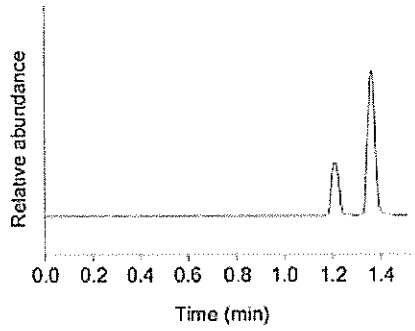
c) Thin-layer chromatography was used to identify which of the three amino acids (A, B or C) was phenylalanine. The chromatogram produced is represented below.



The R_f value of phenylalanine for the conditions used in this separation is 0.75. Identify and explain which of the amino acids **A**, **B** and **C** is phenylalanine. (2 marks)

0.75 × 55 = 39.8 mm (1) so 'B' (1)
or B (1) is more than half, but less than A which would indicate Rf 0.75 (logical answer) (1)

d) Phenylalanine is used in the synthetic production of antidepressants. A sample of phenylalanine was analysed for purity, using a gas chromatograph. The diagram below represents the chromatogram obtained.



State, with a reason, whether or not this sample of phenylalanine is pure. (1 mark)

It is not pure because the chromatogram shows 2 different compounds present.
– need reference to 2 peaks

***Q4.* (11 marks)**

A company that produced a new brand of Apple Juice has employed a chemist to analyse the vitamin C content of their product. The chemist decided to use a redox titration to make the determination. The reaction used was:

$$C_6H_8O_6\ (aq) + I_3^-\ (aq) \rightarrow C_6H_6O_6\ (aq) + 2H^+\ (aq) + 3I^-\ (aq)$$

A 20.00 mL sample of the juice was made up to 250.0 mL with deionised water in a volumetric flask. 25.00 mL aliquots of the diluted juice were then placed in a conical flask and titrated with a 2.00×10^{-4} M I_3^- solution.

The following results were obtained.

Trial	1	2	3	4	5
Titre (mL)	15.87	15.42	15.62	15.67	15.65

a) Calculate the concordant average titre. (1 mark)

$(15.62 + 15.67 + 15.65)/3 = 15.65$ mL

b) Calculate the amount, in mol, of vitamin C present in the 25.00 mL aliquot. (2 marks)

$n(I_3^-) = 0.01565 \times 2.00 \times 10^{-4}$
$= 3.13 \times 10^{-6}$ mol (1)

$n(C_6H_8O_6) = n(I_3^-)$
$n(C_6H_8O_6) = 3.13 \times 10^{-6}$ mol (1)

c) Calculate the concentration of the vitamin C in the original (undiluted) sample of the apple juice in mol L^{-1}. (2 marks)

$n(C_6H_8O_6)$ in flask $= 3.13 \times 10^{-6} \times \frac{250}{25}$
$= 3.13 \times 10^{-5}$ mol (1) $= n(C_6H_8O_6)$ in sample

$[C_6H_8O_6] = \frac{3.13 \times 10^{-5}}{0.02000} = 1.57 \times 10^{-3}$ M (1) -1 if not 3 sig figs.

d) During the analysis the chemist rinsed, but did not dry, each item of glassware. She had available for use: pure water, the original fruit juice, the diluted fruit juice and the standard I_3^- solution. For each item listed below, name the liquid that should be used to rinse it by placing a tick in the appropriate box. (4 marks)

	pure water	original fruit juice	diluted fruit juice	standard I_3^- solution
burette				✓
conical flask	✓			
25.00 mL pipette			✓	
250.0 mL volumetric flask	✓			

(1) each correct answer

e) The chemist's friend suggests that this is probably not the most accurate technique for the determination of the concentration of Vitamin C content. What is a significant error to be considered when conducting a titration like this one. (1 mark)

One of : recognising the correct end point
incomplete reaction
other oxidants

f) The marketers of the Apple juice have stated on the bottle that Vitamin C is an "essential" vitamin. What does this mean? (1 mark)

Humans cannot make it ourselves, it must be in our diet regularly.

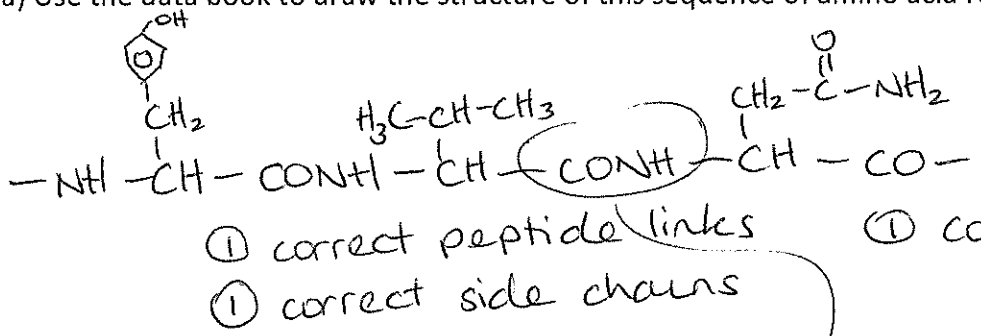
***Q5.* (8 marks)**

The Atlantic longfin inshore squid is able to blend into its surroundings and seemingly disappear. It does this by reflecting light using specialised cells. The squid tunes and adapts to the reflection of light from these cells by using a class of proteins called reflectins.

The amino acid sequences of some reflecting from this squid have been characterised. A small sequence from one of the reflectins is shown below.

-Tyr-Val-Asn-

a) Use the data book to draw the structure of this sequence of amino acid residues of the reflectin. (3 marks)



b) Circle one peptide bond in the structure that you drew in part a). (1 mark)

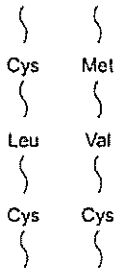
①

c) Describe the α-helix structure of the reflectin protein. (2 marks)

① coiling of the amino acid chain (primary structure)

① caused by H bonding between C=O and N-H in different peptide links.

d) Consider the following amino acids found on neighbouring protein chains as they come into proximity to each other.



Identify the **strongest** interactions that could be found between these neighbouring protein chains and identify **which pairs** they will occur between. (2 marks)

covalent/ disulfide-links (1)
between CYS - CYS (1)

***Q6.* (10 marks)**

Eating a chocolate bar, like a mars bar, provides you with a number of different nutrients. The nutrition information of a Mars bar is shown here.

NUTRITION INFORMATION

SERVINGS PER PACKAGE: 1 SERVING SIZE: 53g

Average Quantities	Per Serving	%DI Per Serve*	Per 100g
ENERGY	1020 kJ	12%	1900kJ
PROTEIN	1.9g	4%	3.6g
FAT – Total	9.1g	13%	16.9g
-Saturated	4.4g	18%	8.2g
CARBOHYDRATE	37.6g	12%	70.3g
- Sugars	30.6g	34%	57.1g
SODIUM	76mg	3%	143mg

a) Write the one or two words needed to answer the following questions and statements about the digestion of the mars bar. (6 marks)

i. Name the process required to digest carbohydrates.	hydrolysis (1)
ii. What part of the digestive system will start the break down proteins?	stomach (1)
iii. What **two** molecules will the fat break down into?	glycerol (1)
	fatty acid (1)
iv. Which nutrient will amylase break down?	starch/complex carb (1)
v. Which essential vitamin could act as an antioxidant to the fats ingested?	Vitamin C (1) (or other correct)

b) The nutrition panel gives the average quantities of carbohydrates and sugars. What is the difference?

(2 marks)

- 'sugar' are a sub-set of carbohydrates eg. mono or disaccharides (1)
- carbohydrates refer to larger class including starch or cellulose (1)

c) Would you expect a mars bar to be high or low GI, compared to pasta? Explain your answer.

(2 marks)

Compare Q – must mention both mars bar and pasta.

- ~~sto~~ GI relates to how fast carbohydrates are absorbed as glucose in the blood (1)
- Mars bars high 'sugar' content means it will have a higher GI than pasta which is starch and would take longer to be absorbed. (1)

End of SAC – Total marks available: 47